

## **networkx**

Used to create and work with graphs

👉 In your project:

Nodes = entities (Fuel, Engine)

Edges = relations (uses, supplies)

👉 This is your **Knowledge Graph engine**

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## **matplotlib**

Used to draw graphs (visualization)

👉 Shows:

Fuel System → Fuel → Combustion

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## **PyPDF2**

Used to read PDF files

👉 Converts:

PDF → Text

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## **scipy**

Used for math operations

👉 Needed for:

PageRank algorithm (important nodes)

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## **numpy**

Used for numerical operations

👉 Helps in:

Calculations inside graph algorithms

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## 🔥 Simple Analogy

networkx = brain (graph)  
matplotlib = eyes (see graph)  
PyPDF2 = reader (read PDFs)  
scipy = calculator (ranking)  
numpy = math helper

This cell installs all required libraries for building, visualizing, and processing the Graph RAG system.

**Next cell :-**

### 1 import os

Used to work with files and folders

👉 In your project:

Check file exists  
Get file names  
Handle paths

Example:

```
os.path.exists("data/file.pdf")
```

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### 2 import re

Used for text cleaning (regular expressions)

👉 In your project:

Remove punctuation  
Clean words  
Extract patterns

Example:

```
re.sub(r'^\w\s', "", text)
```

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### 3 import networkx as nx

Main library to build graph

👉 You will use:

```
nx.DiGraph() # create graph
```

```
graph.add_edge() # add relation
```

👉 nx is just a short name

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🔌 **import matplotlib.pyplot as plt**

Used to draw graph

👉 You will use:

```
plt.figure()
```

```
plt.show()
```

👉 plt = shortcut name

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5 **from PyPDF2 import PdfReader**

Used to read PDF files

👉 You will use:

```
reader = PdfReader("file.pdf")
```

👉 Then:

```
page.extract_text()
```

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🔥 **Simple Understanding**

os → file handling

re → text cleaning

networkx → graph building

matplotlib → graph display

PdfReader → read PDFs

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🧠 **Real Flow Connection**

PDF → (PdfReader)  
↓  
Text → (re cleaning)  
↓  
Triples  
↓  
Graph → (networkx)  
↓  
Visualization → (matplotlib)

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### **One-Line Summary**

This cell imports all necessary libraries required for file handling, text processing, graph creation, and visualization in the Graph RAG system.

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### **Small Note**

You wrote:

```
import
```

👉 That line is incomplete ❌

👉 You can remove it